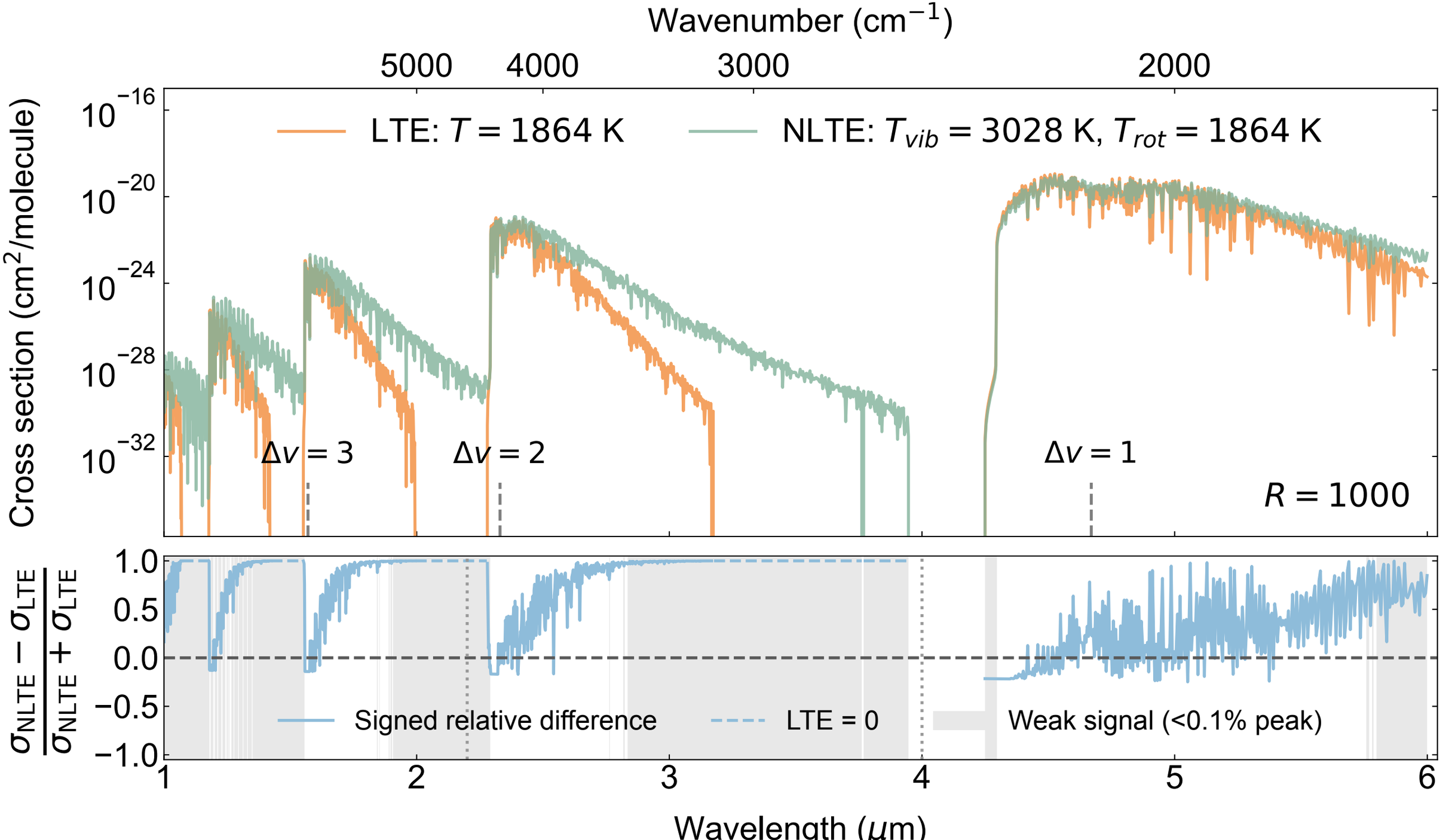

Wavenumber (cm^{-1})
5000
4000
3000
2000
LTE: $T = 1864$ K
NLTE: $T_{vib} = 3028$ K, $T_{rot} = 1864$ K
Cross section (cm^2/molecule)
10^{-16}
10^{-20}
10^{-24}
10^{-28}
10^{-32}
$\Delta v = 3$
$\Delta v = 2$
$\Delta v = 1$
$R = 1000$
$\frac{\sigma_{NLTE} - \sigma_{LTE}}{\sigma_{NLTE} + \sigma_{LTE}}$
1.0
0.5
0.0
−0.5
−1.0
Signed relative difference
LTE = 0
Weak signal (<0.1% peak)
1
2
3
4
5
6
Wavelength (μm)